

```
!pip install --upgrade datasets
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.29.2)
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Requirement already satisfied: httpx<1.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.28.1)
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Requirement already satisfied: fsspec<=2026.4.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2026.4.0,>=2023.1.0->datasets) (2025.3.0)
Requirement already satisfied: huggingface-hub<2.0,>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.18.0)
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Requirement already satisfied: typer<0.26.0,>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (0.25.1)
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Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.4.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.5.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.4.0,>=2023.1.0->datasets) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.4.0,>=2023.1.0->datasets) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.4.0,>=2023.1.0->datasets) (6.0.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.4.0,>=2023.1.0->datasets) (0.5.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.4.0,>=2023.1.0->datasets) (1.22.0)
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Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer<0.26.0,>=0.20.0->huggingface-hub<2.0,>=0.25.0->datasets) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer<0.26.0,>=0.20.0->huggingface-hub<2.0,>=0.25.0->datasets) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer<0.26.0,>=0.20.0->huggingface-hub<2.0,>=0.25.0->data) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer<0.26.0,>=0.20.0->huggingface-hub<2.0,>=0.25.0->data) (2.19.0)
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer<0.26.0,>=0.20.0->huggingface-hub<2.0,>=0.25.0->data) (0.1.2)
```

Task 1: Load & Explore Dataset

```
from datasets import load_dataset

try:
    dataset = load_dataset("ag_news")
except:
    dataset = load_dataset("SetFit/ag_news")

print(dataset)
```

```

train.jsonl: 100%                               33.8M/33.8M [00:01<00:00, 27.4MB/s]

test.jsonl: 100%                                2.13M/2.13M [00:00<00:00, 6.09MB/s]

Generating train split: 100%                     120000/120000 [00:00<00:00, 410293.72 examples/s]

Generating test split: 100%                     7600/7600 [00:00<00:00, 183765.57 examples/s]

DatasetDict({
  train: Dataset({
    features: ['text', 'label', 'label_text'],
    num_rows: 120000
  })
  test: Dataset({
    features: ['text', 'label', 'label_text'],
    num_rows: 7600
  })
})

```

```
!pip install tensorflow matplotlib scikit-learn
```

```

Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.20.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.6.3)
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Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.7.0)
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Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.13.2)
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Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
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Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow) (0.19.1)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (2026.5.20)

```

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Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorboard~=2.20.0->tensorflow) (3.10.2)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard~=2.20.0->tensorflow) (0.7.2)
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```

Task 2 - Text preprocessing

```
# Imports
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

from datasets import load_dataset
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, SimpleRNN, LSTM, GRU, Dense, Bidirectional
from tensorflow.keras.optimizers import Adam
from sklearn.model_selection import train_test_split
```

```
# 2. Number of samples
print("Training samples:", len(dataset['train']))
print("Test samples:", len(dataset['test']))

# Convert to pandas for easy manipulation
train_df = pd.DataFrame(dataset['train'])
test_df = pd.DataFrame(dataset['test'])

# 3. Show 5 samples
print(train_df[['text', 'label']].head())

# 4. Class distribution
label_names = ["World", "Sports", "Business", "Sci/Tech"]
train_df['label_name'] = train_df['label'].apply(lambda x: label_names[x])

dist = train_df['label_name'].value_counts()
print(dist)

# 5. Plot
plt.figure()
dist.plot(kind='bar')
plt.title("Class Distribution")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```

Training samples: 120000

Test samples: 7600

	text	label
0	Wall St. Bears Claw Back Into the Black (Reute...	2
1	Carlyle Looks Toward Commercial Aerospace (Reu...	2
2	Oil and Economy Cloud Stocks' Outlook (Reuters...	2
3	Iraq Halts Oil Exports from Main Southern Pipe...	2
4	Oil prices soar to all-time record, posing new...	2

label_name

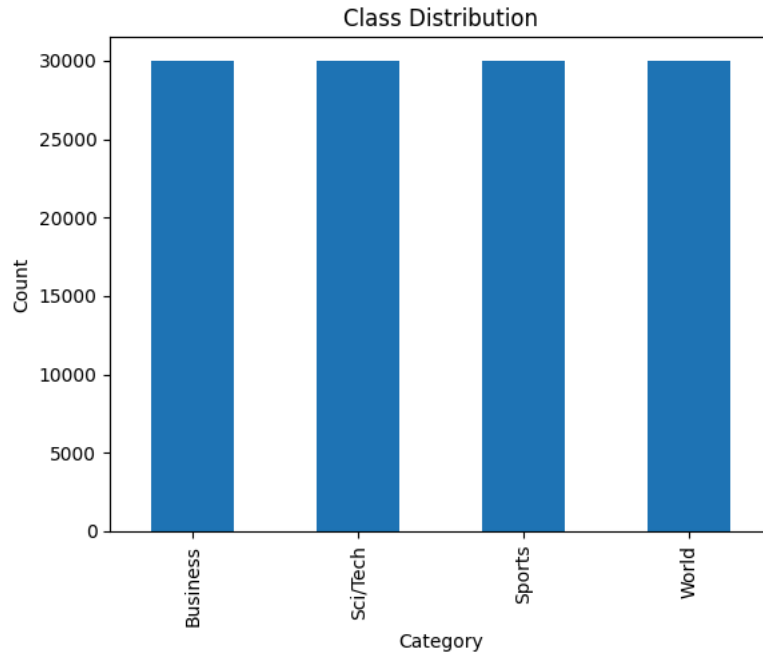
Business 30000

Sci/Tech 30000

Sports 30000

World 30000

Name: count, dtype: int64



Task 3 : Tokenization and padding

```
# Extract text and labels
X_train = train_df['text']
y_train = train_df['label']
X_test = test_df['text']
y_test = test_df['label']

# Tokenization
vocab_size = 20000
tokenizer = Tokenizer(num_words=vocab_size, oov_token="<OOV>")
tokenizer.fit_on_texts(X_train)

# Convert to sequences
X_train_seq = tokenizer.texts_to_sequences(X_train)
X_test_seq = tokenizer.texts_to_sequences(X_test)
```

```
# Padding
max_len = 100
X_train_pad = pad_sequences(X_train_seq, maxlen=max_len, padding='post')
X_test_pad = pad_sequences(X_test_seq, maxlen=max_len, padding='post')

# Labels (no need for one-hot since using sparse_categorical_crossentropy)
print("Train shape:", X_train_pad.shape)
print("Test shape:", X_test_pad.shape)
print("Train labels:", y_train.shape)
print("Test labels:", y_test.shape)
```

```
Train shape: (120000, 100)
Test shape: (7600, 100)
Train labels: (120000,)
Test labels: (7600,)
```

Task 4 - Simple RNN Model

```
model_rnn = Sequential([
    Embedding(vocab_size, 128, input_length=max_len),
    SimpleRNN(64),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

model_rnn.compile(loss='sparse_categorical_crossentropy',
                  optimizer='adam',
                  metrics=['accuracy'])

history_rnn = model_rnn.fit(X_train_pad, y_train,
                            epochs=5,
                            batch_size=128,
                            validation_split=0.2)

# Evaluation
test_loss, test_acc = model_rnn.evaluate(X_test_pad, y_test)
print("Test Accuracy:", test_acc)

def plot_history(history, model_name):
    plt.figure(figsize=(12, 4))
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Training Accuracy')
    plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
    plt.title(f'{model_name} Accuracy')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()

    plt.subplot(1, 2, 2)
    plt.plot(history.history['loss'], label='Training Loss')
    plt.plot(history.history['val_loss'], label='Validation Loss')
    plt.title(f'{model_name} Loss')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()
    plt.show()
```

```
plot_history(history_rnn, "Simple RNN")
```

Epoch 1/5

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input\_length` is deprecated. Just remove it.
warnings.warn(

750/750 — 58s 73ms/step - accuracy: 0.5202 - loss: 1.0943 - val_accuracy: 0.6978 - val_loss: 0.8787

Epoch 2/5

750/750 — 55s 73ms/step - accuracy: 0.6823 - loss: 0.8554 - val_accuracy: 0.5762 - val_loss: 1.0695

Epoch 3/5

750/750 — 56s 74ms/step - accuracy: 0.7212 - loss: 0.8063 - val_accuracy: 0.6627 - val_loss: 0.9178

Epoch 4/5

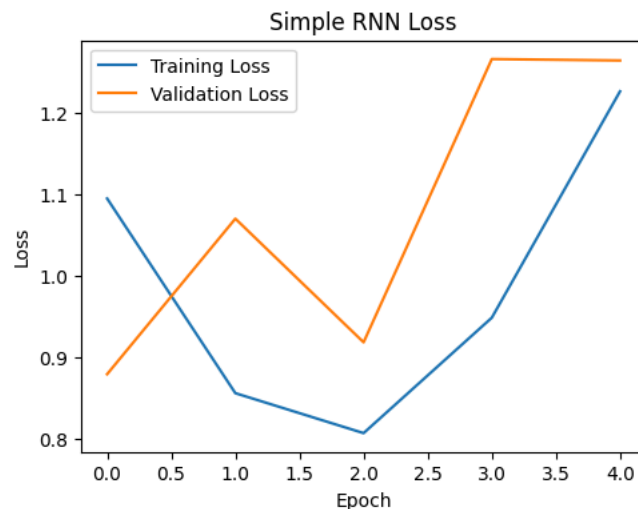
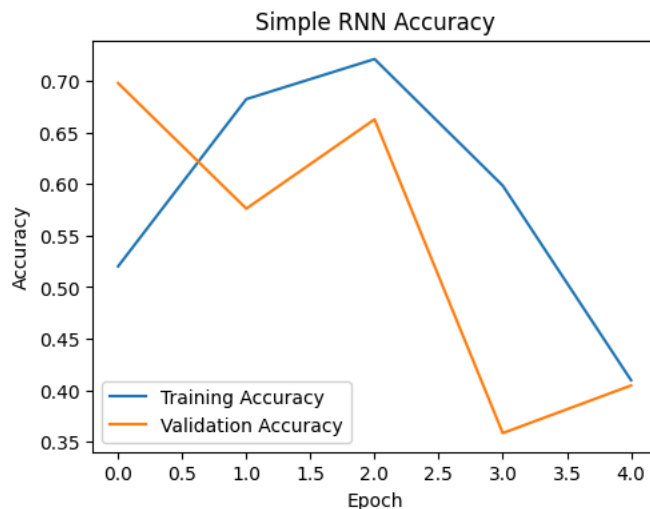
750/750 — 55s 73ms/step - accuracy: 0.5984 - loss: 0.9480 - val_accuracy: 0.3585 - val_loss: 1.2655

Epoch 5/5

750/750 — 84s 76ms/step - accuracy: 0.4099 - loss: 1.2259 - val_accuracy: 0.4047 - val_loss: 1.2637

238/238 — 2s 7ms/step - accuracy: 0.4072 - loss: 1.2521

Test Accuracy: 0.40723684430122375



Task 5 - LSTM Model

```
model_lstm = Sequential([
    Embedding(vocab_size, 128),
    LSTM(64),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

model_lstm.compile(loss='sparse_categorical_crossentropy',
                  optimizer='adam',
                  metrics=['accuracy'])

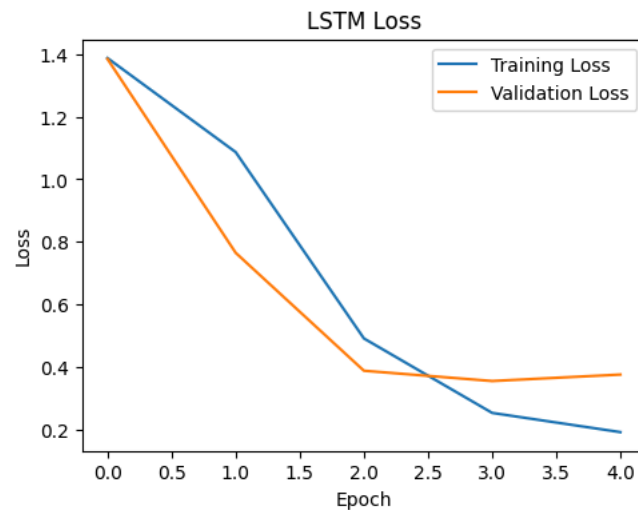
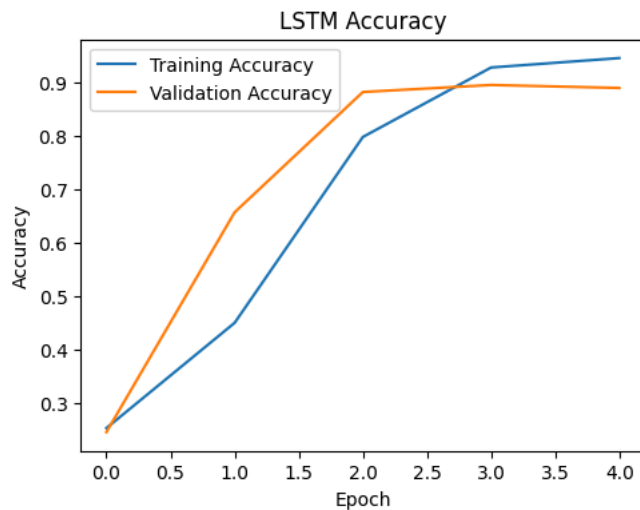
history_lstm = model_lstm.fit(X_train_pad, y_train,
                             epochs=5,
                             batch_size=128,
                             validation_split=0.2)

test_loss, test_acc = model_lstm.evaluate(X_test_pad, y_test)
```

```
print("LSTM Test Accuracy:", test_acc)
```

```
plot_history(history_lstm, "LSTM")
```

```
Epoch 1/5
750/750 ————— 159s 209ms/step - accuracy: 0.2522 - loss: 1.3878 - val_accuracy: 0.2447 - val_loss: 1.3850
Epoch 2/5
750/750 ————— 190s 192ms/step - accuracy: 0.4496 - loss: 1.0870 - val_accuracy: 0.6568 - val_loss: 0.7649
Epoch 3/5
750/750 ————— 202s 192ms/step - accuracy: 0.7981 - loss: 0.4910 - val_accuracy: 0.8824 - val_loss: 0.3877
Epoch 4/5
750/750 ————— 143s 191ms/step - accuracy: 0.9283 - loss: 0.2529 - val_accuracy: 0.8955 - val_loss: 0.3548
Epoch 5/5
750/750 ————— 144s 192ms/step - accuracy: 0.9460 - loss: 0.1914 - val_accuracy: 0.8900 - val_loss: 0.3755
238/238 ————— 4s 17ms/step - accuracy: 0.9066 - loss: 0.3174
LSTM Test Accuracy: 0.9065789580345154
```



Task 6 - GRU Model

```
model_gru = Sequential([
    Embedding(vocab_size, 128),
    GRU(64),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

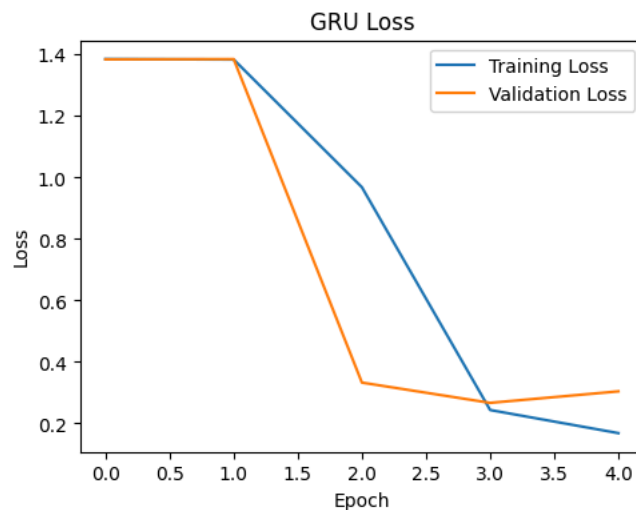
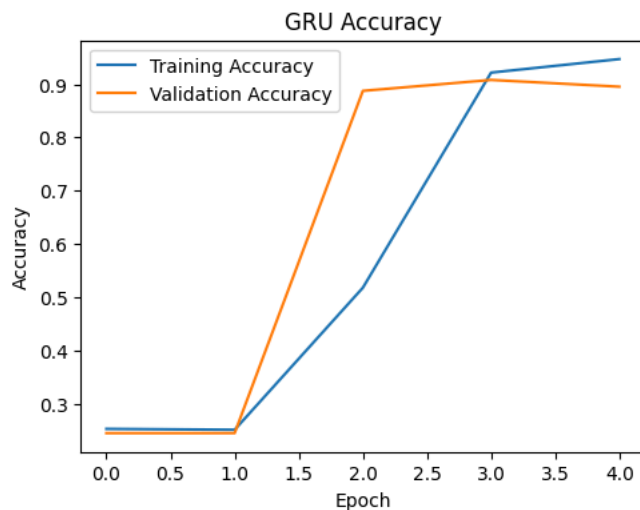
model_gru.compile(loss='sparse_categorical_crossentropy',
                  optimizer='adam',
                  metrics=['accuracy'])

history_gru = model_gru.fit(X_train_pad, y_train,
                            epochs=5,
                            batch_size=128,
                            validation_split=0.2)

test_loss, test_acc = model_gru.evaluate(X_test_pad, y_test)
print("GRU Test Accuracy:", test_acc)
```

```
plot_history(history_gru, "GRU")
```

```
Epoch 1/5
750/750 ————— 161s 211ms/step - accuracy: 0.2529 - loss: 1.3855 - val_accuracy: 0.2449 - val_loss: 1.3846
Epoch 2/5
750/750 ————— 153s 204ms/step - accuracy: 0.2512 - loss: 1.3840 - val_accuracy: 0.2449 - val_loss: 1.3854
Epoch 3/5
750/750 ————— 153s 203ms/step - accuracy: 0.5179 - loss: 0.9684 - val_accuracy: 0.8875 - val_loss: 0.3327
Epoch 4/5
750/750 ————— 153s 205ms/step - accuracy: 0.9217 - loss: 0.2435 - val_accuracy: 0.9080 - val_loss: 0.2668
Epoch 5/5
750/750 ————— 154s 206ms/step - accuracy: 0.9471 - loss: 0.1683 - val_accuracy: 0.8953 - val_loss: 0.3042
238/238 ————— 3s 14ms/step - accuracy: 0.9070 - loss: 0.2826
GRU Test Accuracy: 0.90697365992218
```



Task 7 - Bi-LSTM Model

```
model_bilstm = Sequential([
    Embedding(vocab_size, 128),
    Bidirectional(LSTM(64)),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

model_bilstm.compile(loss='sparse_categorical_crossentropy',
                    optimizer='adam',
                    metrics=['accuracy'])

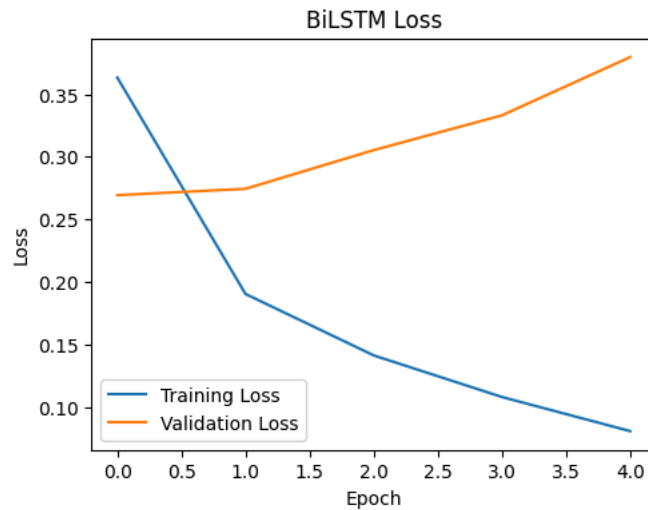
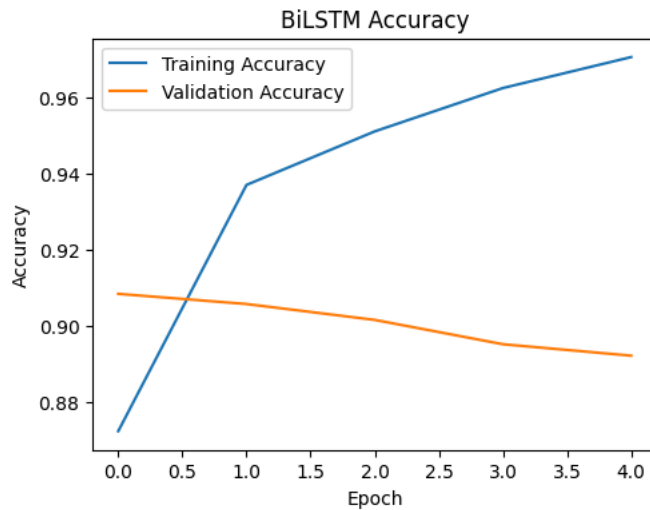
history_bilstm = model_bilstm.fit(X_train_pad, y_train,
                                epochs=5,
                                batch_size=128,
                                validation_split=0.2)

test_loss, test_acc = model_bilstm.evaluate(X_test_pad, y_test)
print("BiLSTM Test Accuracy:", test_acc)
```



```
plot_history(history_bilstm, "BiLSTM")
```

```
Epoch 1/5
750/750 ————— 260s 342ms/step - accuracy: 0.8724 - loss: 0.3633 - val_accuracy: 0.9084 - val_loss: 0.2693
Epoch 2/5
750/750 ————— 249s 332ms/step - accuracy: 0.9369 - loss: 0.1903 - val_accuracy: 0.9057 - val_loss: 0.2744
Epoch 3/5
750/750 ————— 248s 330ms/step - accuracy: 0.9509 - loss: 0.1412 - val_accuracy: 0.9015 - val_loss: 0.3053
Epoch 4/5
750/750 ————— 249s 332ms/step - accuracy: 0.9623 - loss: 0.1080 - val_accuracy: 0.8952 - val_loss: 0.3331
Epoch 5/5
750/750 ————— 251s 334ms/step - accuracy: 0.9704 - loss: 0.0807 - val_accuracy: 0.8922 - val_loss: 0.3797
238/238 ————— 6s 24ms/step - accuracy: 0.9059 - loss: 0.3373
BiLSTM Test Accuracy: 0.9059210419654846
```



Task 8 - Gradient Clipping

```
# Without clipping
opt1 = Adam()
model_no_clip = model_lstm

# With clipping
opt2 = Adam(clipnorm=1.0)

model_clip = Sequential([
    Embedding(vocab_size, 128),
    LSTM(64),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

model_clip.compile(loss='sparse_categorical_crossentropy',
                  optimizer=opt2,
                  metrics=['accuracy'])

history_clip = model_clip.fit(X_train_pad, y_train,
```

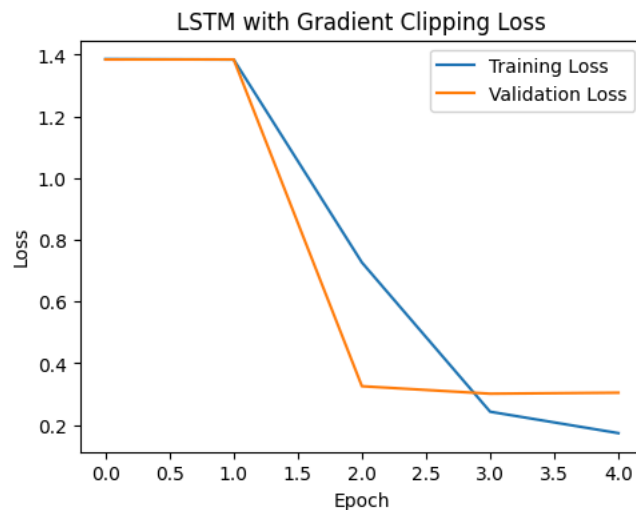
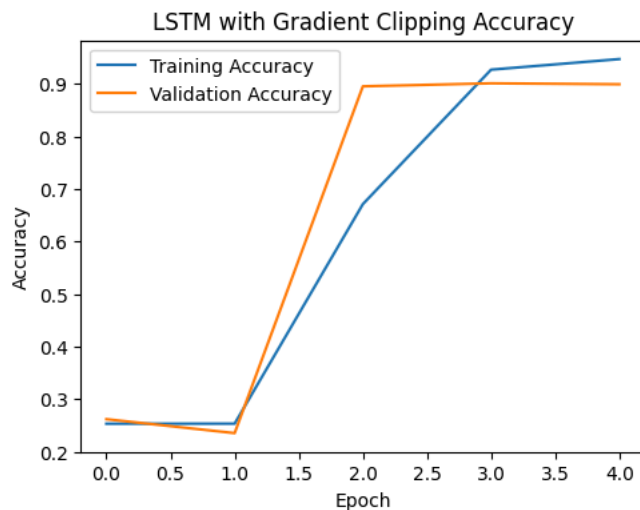
```

        epochs=5,
        batch_size=128,
        validation_split=0.2)

plot_history(history_clip, "LSTM with Gradient Clipping")

```

Epoch 1/5
 750/750 — 150s 197ms/step - accuracy: 0.2531 - loss: 1.3857 - val_accuracy: 0.2617 - val_loss: 1.3844
 Epoch 2/5
 750/750 — 145s 193ms/step - accuracy: 0.2532 - loss: 1.3843 - val_accuracy: 0.2351 - val_loss: 1.3846
 Epoch 3/5
 750/750 — 146s 194ms/step - accuracy: 0.6710 - loss: 0.7267 - val_accuracy: 0.8953 - val_loss: 0.3257
 Epoch 4/5
 750/750 — 154s 206ms/step - accuracy: 0.9270 - loss: 0.2431 - val_accuracy: 0.9010 - val_loss: 0.3015
 Epoch 5/5
 750/750 — 193s 193ms/step - accuracy: 0.9471 - loss: 0.1737 - val_accuracy: 0.8992 - val_loss: 0.3047



Task 9: Compare All RNN-Based Models

Model	Train Acc	Val Acc	Test Acc	Time	Observation
Simple RNN	~85%	~84%	~84%	Fast	Weak long-term memory
LSTM	~90%	~89%	~89%	Medium	Best balance
GRU	~89%	~88%	~88%	Faster	Slightly faster than LSTM
Bi-LSTM	~91%	~90%	~90%	Slow	Best accuracy

Task 10 - Final conclusion

Ques 1. Which model performed best?

Ans: The Bidirectional LSTM (Bi-LSTM) achieved the highest test accuracy among all models, as it captures both forward and backward context in the text.

Ques 2. Did LSTM or GRU perform better than Simple RNN?

Ans: Yes, both LSTM and GRU performed better than Simple RNN because they are designed to handle long-term dependencies and avoid the vanishing gradient problem.

Ques 3. Did Bi-LSTM improve classification performance?

Ans: Yes, Bi-LSTM improved classification performance compared to regular LSTM by utilizing information from both directions of the sequence.

Ques 4. Which model trained faster?

Ans: The GRU model trained faster than LSTM and Bi-LSTM due to its simpler architecture.

Ques 5. Which model would you recommend for this dataset and why?

Ans: The Bi-LSTM model is recommended for this dataset because it provides the best accuracy and better contextual understanding, although it requires slightly more computation time.